

Network Troubleshooting Cheat Sheet

Work from the bottom up. Each step only matters once the one before it passes; the first failure is your problem.

Seven questions, in order

Question	Run	Good sign	If it fails
1. Is the link up?	<code>ip -br link</code>	UP, LOWER_UP	NO-CARRIER = cable / switch port
2. Do I have an IP?	<code>ip -br addr</code>	An IP on main	None, or 169.254.x.x = DHCP failed
3. Can I reach the gateway?	<code>ip route</code> <code>ping -c 3 192.168.1.1</code>	A default via line; replies	No default line = no gateway set. Unreachable = wrong subnet
4. Internet by IP?	<code>ping -c 3 1.1.1.1</code> <code>mtr -n 1.1.1.1</code>	Replies; no loss	Path stops at the gateway = router, NAT or ISP problem
5. Does DNS work?	<code>dig example.com</code> <code>resolvectl status</code>	NOERROR + answer	Fine with dig @1.1.1.1 = your own DNS server is broken
6. Is the port open?	<code>nc -zv example.com 443</code>	succeeded	refused = not listening; timeout = firewall
7. Does it answer?	<code>curl -I https://example.com</code>	HTTP/2 200	4xx/5xx = the app. <code>curl -v</code> shows TLS errors

Reading ping

```
$ ping -c 3 192.168.1.1
64 bytes from 192.168.1.1: icmp_seq=1 ttl=64 time=0.412 ms ← a reply: ttl and round-trip time
64 bytes from 192.168.1.1: icmp_seq=2 ttl=64 time=0.389 ms
3 packets transmitted, 3 received, 0% packet loss, time 2003ms ← any loss = trouble
rtt min/avg/max/mdev = 0.389/0.400/0.412/0.009 ms ← mdev = jitter (how much it varies)
```

You see	Means	You see	Means
time=	Round trip (LAN under 1 ms)	Host Unreachable	No device on the LAN has that IP
ttl=	Hops left (starts at 64 on Linux)	No output at all	Host down, or ping is blocked
icmp_seq gaps	Those packets were lost	...not known	DNS failed, not the network

Reading dig

```
$ dig example.com
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 41237 ← status: did the lookup work?
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1 ← ANSWER: 0 = no record
;; ANSWER SECTION:
example.com.      3600    IN      A       203.0.113.10 ← name, seconds to cache, type, the answer
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP) ← which DNS server replied
```

Status	Means	Status	Means
NOERROR	Worked (check ANSWER is not 0)	SERVFAIL	The DNS server couldn't get an answer
NXDOMAIN	Name doesn't exist (typo?)	timed out	Can't reach the DNS server at all

Handy recipes

Command	Does
<code>dig +short example.com · dig -x 203.0.113.10</code>	Just the IP · reverse lookup (IP to name)
<code>traceroute -n example.com</code>	Each router on the way; * * * = hop didn't answer (often fine)
<code>curl -s -o /dev/null -w '%{http_code}\n' URL</code>	Prints only the HTTP status code
<code>ip neigh</code>	Neighbors on the LAN; FAILED = no reply from that IP

Remember: ping by IP works but by name fails = **DNS**. Connection **refused** = you reached the host but nothing is listening; **timeout** = a firewall is silently dropping it. Some hosts block ping, so a failed ping alone isn't proof.